

Dynamic Hand Gesture Recognition Using Improved Spatio-Temporal Graph Convolutional Network

Jae-Hun Song[✉], Kyeongbo Kong[✉], and Suk-Ju Kang[✉], *Member, IEEE*

Abstract—Hand gesture recognition is essential to human-computer interaction as the most natural way of communicating. Furthermore, with the development of 3D hand pose estimation technology and the performance improvement of low-cost depth cameras, skeleton-based dynamic hand gesture recognition has received much attention. This paper proposes a novel multi-stream improved spatio-temporal graph convolutional network (MS-ISTGCN) for skeleton-based dynamic hand gesture recognition. We adopt an adaptive spatial graph convolution that can learn the relationship between distant hand joints and propose an extended temporal graph convolution with multiple dilation rates that can extract informative temporal features from short to long periods. Furthermore, we add a new attention layer consisting of effective spatio-temporal attention and channel attention between the spatial and temporal graph convolution layers to find and focus on key features. Finally, we propose a multi-stream structure that feeds multiple data modalities (i.e., joints, bones, and motions) as inputs to improve performance using the ensemble technique. Each of the three-stream networks is independently trained and fused to predict the final hand gesture. The performance of the proposed method is verified through extensive experiments with two widely used public dynamic hand gesture datasets: SHREC'17 Track and DHG-14/28. Our proposed method achieves the highest recognition accuracy in various gesture categories for both datasets compared with state-of-the-art methods.

Index Terms—Hand skeleton, graph convolutional network, graph attention, multi-stream network.

I. INTRODUCTION

HAND gestures are body language that enable natural communication through the center of the palm and the finger position. They are classified into static and dynamic

gestures [1], [2]. A static hand gesture is represented by the stable shape of the hand from a single image of stationary gestures, whereas a dynamic hand gesture additionally includes movement from a sequence of a hand such as waving. Dynamic hand gesture recognition is becoming increasingly crucial and used in various applications such as sign language communication [3], human-computer interaction [4], virtual reality [5], [6], robot control [7], [8], and game control [9].

The primary approaches to hand gesture research can be divided into the wearable glove-based sensor approach and the vision-based approach [10]. Glove-based methods [11] using gloves to obtain hand movement information cannot be widely used because they require expensive hardware equipment and have restrictions in wearing them directly. Therefore, studies on the natural vision-based method [12]–[35] have been active because they are relatively inexpensive and do not require equipment. The hand gesture data used in vision-based methods can be divided into two modalities: image-based (e.g., RGB videos or depth image sequences) and skeleton-based. Compared with image-based methods [12]–[16], skeleton-based methods [17]–[35] have the advantage of being more discriminative and biologically informative in expressing hand gestures. They are also robust to various illuminations and complex backgrounds. Because the position of the joint is expressed in 3D coordinates, its data size is smaller, and its computational efficiency is higher than RGB or depth images.

Furthermore, the wide use of advanced hardware devices such as Microsoft Kinect [36] and Intel RealSense [37] has recently made it easier to obtain depth sequences. The continuous development of hand pose estimation algorithms [38]–[42] to predict 3D coordinates from acquired depth images has enabled accurate hand skeleton sequences to be generated in real-time. Due to these advantages, skeleton-based hand gesture recognition has become increasingly popular and widely used in recent research. This paper focuses on this skeleton-based dynamic hand gesture recognition task.

Skeleton-based dynamic hand gesture recognition is still a challenging task, to which the feature extraction method is critical. Feature extraction can be divided into two main methods: handcrafted features and deep learning-based features. Conventional skeleton-based gesture recognition methods analyze spatial features through handcrafted features [17]–[21]. However, these handcrafted features cannot efficiently model spatial and temporal meaningful information. Methods using deep learning-based features [22]–[35] feed hand skele-

Manuscript received 5 January 2022; revised 15 March 2022; accepted 29 March 2022. Date of publication 5 April 2022; date of current version 6 September 2022. This research is supported by Ministry of Culture, Sports and Tourism and Korea Creative Content Agency (Project Number: R2020040058), and the Korea Agency for Infrastructure Technology Advancement (KAIA) grant funded by the Ministry of the Interior and Safety (Grant 22PQWO-C153369-04), and Convergent Technology R&D Program for Human Augmentation through the National Research Foundation of Korea (NRF) funded by Ministry of Science and ICT (2020M3C1B8081320). This article was recommended by Associate Editor C. Chen. (*Corresponding author: Suk-Ju Kang.*)

Jae-Hun Song and Suk-Ju Kang are with the Vision and Display Systems Laboratory for Electronic Engineering, Sogang University, Seoul 04017, South Korea (e-mail: songjh1@sogang.ac.kr; sjkang@sogang.ac.kr).

Kyeongbo Kong is with the Media School, Pukyong National University, Busan 48547, South Korea (e-mail: kbkong@pknu.ac.kr).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2022.3165069>.

Digital Object Identifier 10.1109/TCSVT.2022.3165069

1051-8215 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

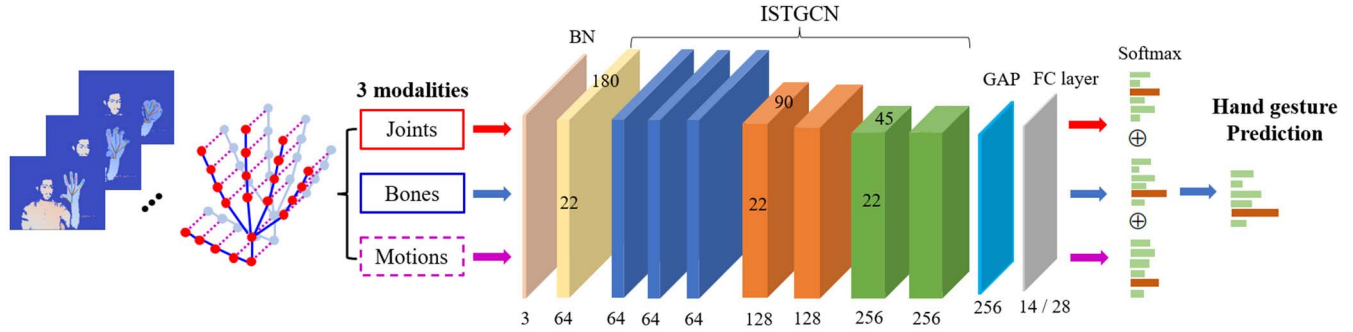


Fig. 1. Overall architecture of the proposed method with multi-stream improved ST-GCN. This architecture consists of ISTGCN blocks, and the marked number denotes the feature dimensions. There are three modalities fed into the network separately: joints, bones, and motions. Each modality corresponds to one stream. The final gesture is predicted as the weighted summation of the three softmax scores.

ton sequences to recurrent neural networks (RNNs) and convolutional neural networks (CNNs) networks for hand gesture recognition. However, these methods cannot fully exploit spatial connection information between joints in each frame and temporal connection information between joints in consecutive frames.

Recently, graph convolutional networks (GCNs) among deep learning-based methods have been widely used to learn high-level graph structure features and have exhibited high performance. Yan *et al.* [43] first proposed a spatio-temporal GCN (ST-GCN) that extracts the two features (spatial and temporal) by applying GCN for the skeleton-based human action recognition task. Since then, several studies have applied this method to action recognition task [44]–[52] and dynamic hand gesture recognition task [53], [54]. It is also applied to application tasks such as driver behavior recognition [55].

However, those previous studies are insufficient for the hand gesture recognition task because pre-fixed graphs based on the hand skeleton's physical connection structure are used for spatial information extraction. Furthermore, because these ST-GCN methods only use the original 3D coordinates of the joint as inputs, motion information such as the trajectory of the hand and the length or direction features of the bones between the two joints cannot be captured.

This paper proposes a novel improved spatio-temporal graph convolutional network for dynamic hand gesture recognition. Fig. 1 illustrates the overall architecture of the proposed method. We adopt a multi-stream structure that uses joint, bone, and joint motion information as inputs for skeleton-based dynamic hand gesture recognition. The fusion of three-stream networks demonstrates a significant performance improvement compared with the previous single input stream method. These stream networks are independently trained using the same network architecture and fused to predict the hand gesture class. We increase the model's flexibility by adopting an adaptive graph convolution layer to extract informative spatial features from different samples. We then apply an attention mechanism to find and selectively focus on key differential information on the activation of joints, frames, and channels. Furthermore, multiple-scale temporal modeling is applied using temporal convolutions with different

dilation rates to extract temporal features from gestures of different lengths. We demonstrate the superiority of the proposed architecture by performing extensive experiments on two widely used public benchmark datasets: SHREC'17 Track and DHG-14/28.

The main contributions of this paper are as follows:

- 1) We propose a novel ST-GCN block that extracts the enhanced spatial and temporal features. In each block, the adaptive spatial graph convolution layer extracts improved spatial features compared with the static graph containing only limited connection information, and the temporal convolution layer with different dilation rates extends the receptive field to extract more informative temporal features
- 2) The newly proposed spatial and temporal co-attention increases the representation of features by jointly considering each dependency as a pair of spatial and temporal directions. Moreover, we apply channel attention together to improve performance further.
- 3) We further improve performance using low-level features by proposing, for the first time, a multi-stream network using three modalities—joints, bones, and motions—for a hand gesture recognition task.
- 4) We conducted extensive ablation experiments on skeleton-based dynamic hand gesture recognition and action recognition datasets. Experimental results on public hand gesture recognition datasets show that the proposed method achieves better performance than previous state-of-the-art methods.

The rest of the paper is organized as follows. Section II reviews the related work. The proposed multi-stream network is described in Section III, and extensive experimental results are presented in Section IV. Finally, Section V concludes the paper.

II. RELATED WORK

Some representative and related work on dynamic hand gesture recognition are introduced in this section. We discuss the latest work which apply attention mechanism in dynamic hand gesture recognition. Furthermore, we review recent studies extracting spatial features using GCNs.

A. Skeleton-Based Dynamic Hand Gesture Recognition

As previously introduced, dynamic hand gesture recognition can be divided into two categories according to features: handcrafted feature-based methods and deep learning-based methods.

Handcrafted feature-based methods [17]–[21] encode hand skeleton sequences into handcrafted features. Smedt *et al.* [17] designed effective descriptors using the hand joints provided by the depth camera as a representative method. These handcrafted descriptors represent the geometric shape of the hand and the combined features of the wrist rotation and hand direction. Each descriptor is encoded in a Fisher vector—an image representation obtained by pooling local image features using the Gaussian mixture model (GMM). A temporal pyramid is used to encode temporal features, and a linear support vector machine (SVM) classifier is applied in the final step to classify the gesture class.

Deep learning-based methods [22]–[35] encode the hand skeleton sequences into coordinate vectors and feed them into RNNs, CNNs, and GCNs to extract spatial and temporal features for gesture recognition. Chen *et al.* [22] proposed a model to learn the motion features of hand gestures using an RNN. This model used two features to represent finger movements of a hand skeleton and global movements of rotation and translation, respectively. A bidirectional RNN was then applied to recognize hand gestures. Nunez *et al.* [27] arranged hand skeleton sequences as pseudo images. Frames correspond to columns of the image, joints to rows, and 3D coordinates to its three channels. Furthermore, this model applied a 2D CNN to extract spatial and temporal features for final gesture classification. However, these methods cannot effectively use spatio-temporal contexts such as connection information between hand joints in each frame and hand joint connection information between frames. Nguyen *et al.* [32] used manifold learning to improve gesture recognition. Low-dimensional symmetric positive definite (SPD) matrices learned from skeleton data are more discriminative. However, this method models skeleton sequences in a predefined structure. Therefore, it cannot capture efficiently the features of joint relationships.

In contrast, GCN-based methods can exploit this information. Li *et al.* [54] first applied an ST-GCN [43] for a human action recognition task to a dynamic hand gesture recognition task. This model established three types of additional edges and fixed edges by kinematic links to capture the relationship between non-adjacent joints more precisely. However, because the form of the spatial graph was fixed, it was not practical for handling various spatial joint relationships for different hand gestures. In addition, it could not model the temporal joint connections between frames sufficiently due to the use of a fixed temporal dimension.

B. Skeleton-Based Human Action Recognition

The skeleton-based action recognition task is similar to the skeleton-based hand gesture recognition task. Several methods of skeleton-based action recognition can be generalized to hand gesture recognition. In this section, we review

GCN-based methods, which are being actively studied among deep learning-based methods. Yan *et al.* [43] first embedded a skeleton sequence into a spatio-temporal graph. In their graph, the spatial edge was fixed by the natural human skeleton, and the temporal edge connected two joints at the same position between consecutive frames. ST-GCN was then proposed to extract spatial and temporal features of the skeleton sequence for action class recognition. Since that study, various GCN-based methods [44]–[56] have been proposed to improve performance. Shi *et al.* [44] proposed a two-stream adaptive GCN (2s-AGCN) that can adaptively learn graph topology. In addition to the first-order information of original joint coordinates, the length and direction of the bone were used as second-order information of the joint data. Liu *et al.* [49] proposed a unified spatio-temporal convolution (G3D) and multi-scale graph convolution network. The proposed multi-scale aggregation method disentangled the importance of nodes in different neighborhoods for effective long-range modeling. Zhang *et al.* [50] adopted a data preprocessing strategy, whose inputs include position, velocity, joint type, and frame index. This data preprocessing module can significantly reduce the model size, but its performance is lower than other methods. Shi *et al.* [45] applied an ensemble of four models, trained with the joint, bone, joint motion, and bone motion features. Although these methods achieve significant accuracy, the increased computational cost due to the multi-stream structure and 3D convolution hinder practical application. Therefore, there is a need for a smaller-sized single stream and an efficient multi-stream network which is fusing optimal input features.

C. Attention Mechanism

Recently, the attention mechanism is widely used in various tasks. Some works [35], [53], [56] have been applying the attention mechanism for dynamic hand gesture recognition. Hou *et al.* [35] proposed a model that learns different attention levels. They assigned each level to a spatio-temporal feature extracted by the convolution network. This model applied the mask branch to focus adaptively on informative time frames and features. Chen *et al.* [53] constructed dynamic graphs using spatial-temporal attention. They applied the self-attention mechanism that is performed in both spatial and temporal domains.

With respect to action recognition, Si *et al.* [56] proposed an attention-enhanced graph convolutional long short-term memory network (AGC-LSTM) to select discriminative spatial information. Shi *et al.* [45] proposed an attention mechanism consisting of three parts: spatial, temporal, and channel attention modules. The attention mechanism is used to enhance the information of key joints in each layer.

Compared with the previous works that apply attention mechanism on spatial domain or temporal domain separately, we use the spatial and temporal co-attention to obtain more discriminative features in GCN for the dynamic hand gesture recognition.

D. Spatial Graph Convolution Network

The input of traditional CNNs is usually Euclidean data in the form of regular grids such as images or videos.

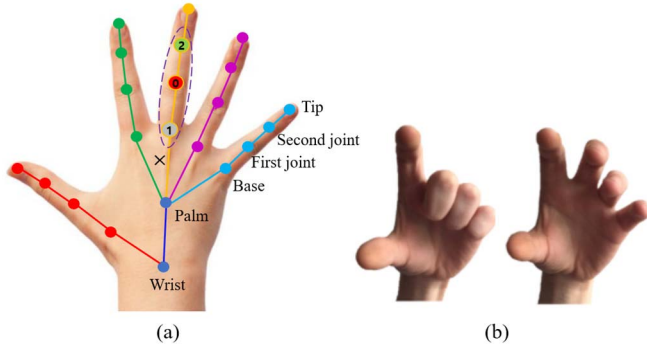


Fig. 2. SHREC'17 Track [61] and the DHG-14/28 [17] collected in the same way: (a) Hand skeleton with 22 hand joints. The cross marks the center of gravity. And the three subsets are divided into a root node (red color) and a centripetal group (gray color) closer to the root node and a centrifugal group (green color) farther than the root node. (b) Two hand shapes (one finger and the whole hand).

However, because the graph is irregular in shape and size, it is not suitable to model graph data because CNNs use a grid kernel. Hand gesture skeleton data comprise a graph structure, and an adjacency matrix or a Laplacian matrix [57] is used to analyze the features more accurately.

The spatial skeleton graph can be represented by $G = \{V, E\}$, where V is the vertex (node) set, and E is the edge set. The spatial graph convolution network aggregates information from the node features of the graph. The features of adjacent nodes are aggregated by an aggregation function. Therefore, the convolution of the spatial graph is calculated according to the neighborhood of nodes—it is important to select the neighborhood nodes of the graph. We adopted the spatial partition strategy proposed by Yan *et al.* [43] of dividing neighborhood nodes into three subsets. As shown in Fig. 2(a), three subsets consist of the root node itself, centripetal group, and centrifugal group. The center of gravity was first generated by calculating the average coordinates of all hand joints. With this center of gravity as a reference point, the centripetal group contains the neighborhood nodes close to the center of gravity, and the centrifugal group contains the neighborhood nodes farther away from the center of gravity.

The adjacency matrix and degree matrix are denoted by A and D , expressed as follows:

$$D(i, i) = \sum_j A(i, j) + \alpha, \quad (1)$$

where $A(i, j)$ indicates whether nodes of i and j are connected. $D(i, i)$ is the diagonal node degree matrix, and α is set to 0.001 to avoid zero.

The adjacency matrix A is normalized by the degree matrix D according to [57] as follows:

$$A_{norm} = D^{-1/2} A D^{-1/2}. \quad (2)$$

The spatial convolution operation can be expressed as an ST-GCN [43]:

$$f_{out}(v_i) = \sum_{v_j \in B_i} \frac{1}{Z_{ij}} f_{in}(v_j) \cdot w(l_i(v_j)), \quad (3)$$

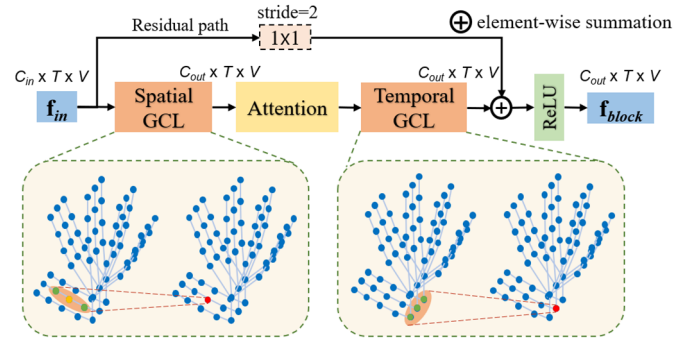


Fig. 3. Structure of the ISTGCN block. The block comprises a spatial graph convolution layer, attention layer, and temporal graph convolution layer—with a residual connection for each block, except for first block. The first embedding block for embedding 3-channel input data into 64-channel features does not apply the residual connection.

where $f_{in}(v_j)$ is the feature map for node v_j and B_i is the neighborhood set for node v_i . w is the weighting function similar to the convolution operation that calculates a weight vector.

III. PROPOSED METHOD

This section presents the architecture and components of our proposed MS-ISTGCN in detail.

A. Network Architecture

The overall proposed architecture is depicted in Fig. 1, where the basic structure is designed based on an ST-GCN [43]. The network comprises eight improved spatio-temporal graph convolution blocks connected in series (refer to Section IV). Based on the settings in [27], the input data with the size of $C_{in} \times T \times V$ is normalized by the batch normalization (BN) layer for the first time. C_{in} is the number of channels, and the 3D coordinates of joints are the input, so its value is 3. T is the number of frames in each sequence, and V is the number of joints of hand skeleton data.

The input data are fed to the ISTGCN block as depicted in Fig. 3, and high-level features are gradually extracted through eight blocks. Each ISTGCN block consists of three layers connected in the order of a spatial graph convolution layer (GCL), an attention layer, and a temporal GCL. The blocks have 64, 64, 64, 64, 128, 128, 256, and 256 output channels, respectively. The convolution stride of the fifth and seventh blocks is set to 2 to reduce the length of the frame sequence. The output feature of the last eighth block is averaged as a feature map with 256 channels using a global average pooling (GAP) layer. The fully connected (FC) layer and softmax are then used to obtain the final class prediction.

B. Spatial Graph Convolution Layer

The GCN-based recognition task's graph topology—fixed by depending only on physical connections—reflects insufficient latent information between skeletons. Therefore, some

previous studies have tried to design an optimal graph topology. As introduced in Section II, ST-GCN [43] transforms Equation (3) into the following:

$$f_{out} = \sum_k^{K_v} W_k f_{in} (A_k \otimes M_k), \quad (4)$$

where K_v is the kernel size of the spatial dimension and the number of subsets is 3 due to the spatial partition strategy. W_k is a weight vector of a 1×1 convolution operation. A_k is the normalized adjacency matrix expressed in Equation (2). For each adjacency matrix A_k , we use a $V \times V$ learnable attention map M_k to represent the importance of each edge. $\otimes$ denotes element-wise multiplication.

In our network, Equation (3) is transformed into the following:

$$\alpha \cdot f_{out} = \sum_k^{K_v} W_k f_{in} (A_k + B_k + C_k), \quad (5)$$

where A_k is the normalized adjacency matrix. B_k is an adaptive graph for global attention with all values parameterized, and it functions similar to M_k in the ST-GCN. C_k is a sample individual adaptive graph representing the similarity that determines the connection strength of two joints in the embedded space. The normalized Gaussian function calculated by the softmax operation, used in the non-local structure, is applied to calculate similarity C_k . A parameterized coefficient α is multiplied with C_k to adjust the importance of the individual graph for each ISTGCN block.

Based on the successful performance of the adaptive GCN [44] for action recognition, we construct a spatial graph convolution layer for dynamic hand gesture recognition as depicted in Fig. 4(a). The output of the spatial graph convolution layer can be formulated as follows:

$$f_{spatial} = ReLU [BN (f_{out}) + I (f_{in})], \quad (6)$$

where BN is the batch normalization, and I is the identity mapping for stability during training. $ReLU$ is then applied as an activation function.

C. Extended Temporal Convolution Layer

Previous studies including ST-GCN [43], [44], [54], aggregate temporal features by performing $\Gamma \times 1$ convolution with one temporal kernel size (Γ) throughout the architecture. Therefore, the sampling range $R(v_t)$ for the temporal dimension can be written as:

$$R(v_t) = \left\{ v_q \mid q - t \leq \left\lfloor \frac{\Gamma}{2} \right\rfloor \right\}, \quad (7)$$

where v_t represents the joints in frame t . Previous studies typically use a 9×1 kernel, which aggregates information from the joints of each of the four frames before and after the target joint.

We design a structure that can perform multiple-scale temporal aggregation to model hand gestures of different lengths. As depicted in Fig. 4(b), the proposed extended temporal graph convolution layer consists of four branches. We use

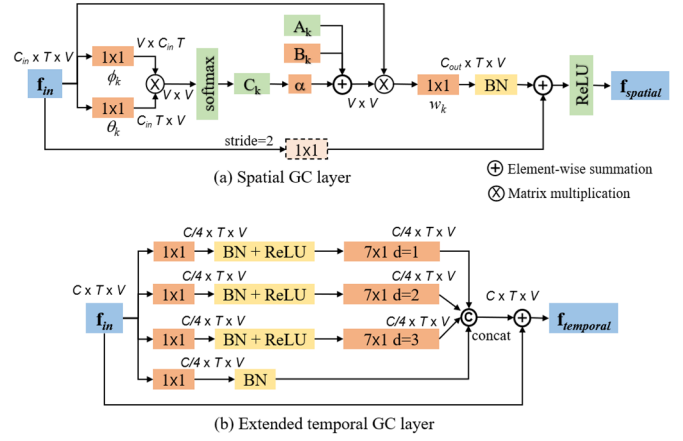


Fig. 4. Architecture of our spatial graph and temporal graph convolution layers: (a) adaptive spatial graph convolution (GC) layer and (b) extended temporal graph convolution (E-TGC) layer. ‘ 1×1 ’ block denotes 1×1 convolution operation and the orange block indicates that the part is the learnable parameter. The term “d” of 7×1 convolutions in temporal GC layer is “dilation.”

a bottleneck structure [58] with 1×1 convolution to reduce the computational cost caused by the branches. The number of channels in each branch is set to the number of input channels divided by four (the number of branches). Thus, the number of channels in the concatenated feature is equal to the number of input channels. We extend the receptive field by applying three different dilation rates ($d = 1, 2, 3$) to three temporal convolutions with a kernel size of 7×1 . The features of the four branches are concatenated, and a residual path is added to facilitate training. The proposed method refers to the multi-scale graph convolution network proposed by Liu *et al.* [49], but it is optimized in terms of performance and model size for the hand gesture recognition task. Unlike the structure of [49], we simplify the number of branches of the temporal GC layer from six to four, and the number of temporal GC layers is reduced from three to one by applying convolutions with the optimal kernel size.

D. Graph Attention Layer

Recently, the application of an attention mechanism in a convolution block has received much attention and has been used in many studies [45], [46], [59], [60]. Squeeze-and-Excitation Network (SENet) [59] is a representative algorithm that has significantly improved the performance of various deep CNN architectures. Studies on various types of attention structures [46], [60] have been actively conducted.

Several methods have been proposed for skeleton-based action recognition. Shi *et al.* [45] sequentially applied attention to each spatial, temporal, and channel dimension. Recently, Song *et al.* [46] proposed an attention method for joint spatio-temporal information inspired by coordinate attention [60] for CNNs.

The attention layer of [46] is a structure in which the channel dimension is reduced and increased again using two FC layers, such as SENet, to perform the spatio-temporal joint attention and channel attention together. However, a loss of

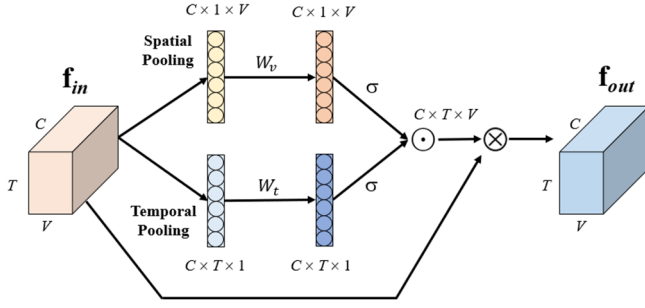


Fig. 5. Effective spatio-temporal co-attention (Eff-ST attention) layer. W_v and W_t are 1×1 convolutions. $\odot$ denotes the outer product and $\otimes$ denotes the element-wise multiplication.

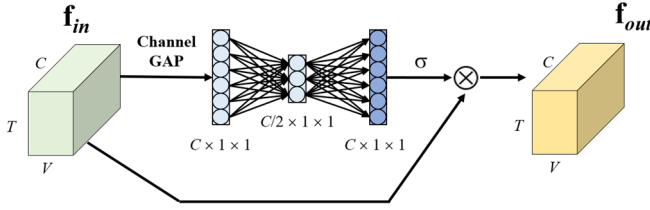


Fig. 6. Channel attention layer consisting of two FC layers. $\otimes$ denotes the element-wise multiplication.

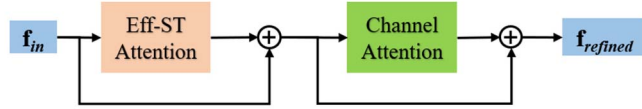


Fig. 7. Proposed attention layer consisting of two sub-layers. A residual connection is respectively added for each sub-layer to stabilize the training.

spatio-temporal information occurs due to channel reduction. Therefore, we introduce an effective spatio-temporal attention (Eff-ST) layer without loss of information while preserving co-attention of the spatio-temporal dimension, as depicted in Fig. 5. This novel attention layer does not perform channel reduction after average pooling. One 1×1 convolution is applied for each spatial and temporal direction to prevent information loss. Furthermore, the channel attention layer used in [45] is applied to emphasize the key information on the channel dimension, as depicted in Fig. 6. Therefore, we design effective attention layers composed of two sub-layers by adding them as a series, as depicted in Fig. 7.

The formula of the effective spatio-temporal co-attention (A_{st}) is as follows:

$$A_{st} = \sigma(AvgPool_v(f_{in}) \cdot W_v) \odot \sigma(AvgPool_t(f_{in}) \cdot W_t), \quad (8)$$

$$f_{out} = A_{st} \otimes f_{in} \oplus f_{in}, \quad (9)$$

where f_{in} and f_{out} are input and output features, respectively. $AvgPool$ is average pooling. $\odot$ and $\otimes$ denote channel-wise outer-product and element-wise multiplication, respectively. $\oplus$ is for element-wise summation. σ is Sigmoid activation function, $W_v \in \mathbb{R}^{C \times C}$ and $W_t \in \mathbb{R}^{C \times C}$ are the learnable weights of two convolutions, respectively.

The channel attention layer generates the channel weights using two nonlinear FC layers and a sigmoid activation function after GAP for the channel. The formulation of the channel attention (A_c) layer is as follows:

$$A_c = \sigma((ReLU(GlobalAvgPool_c(f_{in}) \cdot W_1)) \cdot W_2), \quad (10)$$

$$f_{out} = A_c \otimes f_{in} \oplus f_{in}, \quad (11)$$

where W_1 and W_2 are the weights of two FC layers, respectively.

Algorithm 1 presents the pseudocode for the proposed attention layer.

Algorithm 1 Proposed Eff-ST and Ch Attention Layers

```

def attention_layer(x): # input_feature x (C, T, V)
# Note: Pseudo code for Attention Layers # (channels,
frames, joints)
X_s = Avg_Pool(x) # shape = (C, I, V)
X_s = Conv_v(X_s)
X_s = Sigmoid(X_s) # shape = (C, I, V)
X_t = Avg_Pool(x) # shape = (C, T, I)
X_t = Conv_t(X_t)
X_t = Sigmoid(X_t) # shape = (C, T, I)
X = OuterProduct(X_s, X_t) # shape = (C, T, V)
X = Elementwise_Mul(x, X) # Output of Eff-ST attention
Y_in = x + X # Input of Channel attention
Y_ch_pool = Global_Avg_Pool(Y_in) # shape = (C, I, I)
Y_ch_pool = FC1(Y_ch_pool) # shape = (C/2, I, I)
Y_ch_pool = ReLU(Y_ch_pool)
Y_ch_pool = FC2(Y_ch_pool) # shape = (C, I, I)
Y_ch_pool = Sigmoid(Y_ch_pool) # shape = (C, I, I)
Y = Elementwise_Mul(Y_in, Y_ch_pool) # shape = (C, T, V)
Y = Y_in + Y
return Y # output_feature (C, T, V)

```

E. Multi-Stream Network

We design a multi-stream network using three different data modalities, as depicted in Fig. 1. We use an ensemble strategy of fusing after training each of the three networks with the same architecture. In addition to the original joint coordinates provided by the datasets, the length and direction information of the bones and joints motion information can further improve the performance of skeleton-based recognition tasks. The three modalities we use as inputs to each stream network are joints, bones, and motion, respectively. Joint coordinates can be represented as follows:

$$J_{i,t} = (x_{i,t}, y_{i,t}, z_{i,t}), \forall i \in V, t \in T \quad (12)$$

For example, as shown in Fig. 2(a), the dynamic hand gesture recognition dataset uses 22 joints, and each finger has 4 joints. The length of the bone is calculated as the difference between the two adjacent joints, and the direction of the bone is away from the center of gravity of the joints. In Fig. 2(a), dots and solid lines represent joints and bones, respectively. Given the source joint $J_{i,t}$ close to the center of gravity and

the target joint $J_{j,t}$ farther away from the center of gravity, the bone data can be calculated as follows:

$$B_{i,j,t} = (x_{j,t} - x_{i,t}, y_{j,t} - y_{i,t}, z_{j,t} - z_{i,t}). \quad (13)$$

Moreover, motion can be calculated by the coordinate difference between the same joints in two consecutive frames:

$$M_{i,t} = J_{i,t+1} - J_{i,t}, \forall i \in V, t \in T \quad (14)$$

Finally, the softmax scores of the three streams are fused using weighted summation, and the hand gesture class is predicted based on this score.

IV. EXPERIMENTS

In this section, we evaluate the performance of the proposed MS-ISTGCN and conventional methods with two widely used public skeleton-based dynamic hand gesture datasets: SHREC'17 Track [61] and DHG-14/28 [17]. Furthermore, we perform various ablation studies to verify the effectiveness of each component of the proposed model.

A. Datasets

1) *SHREC'17 Track*: The SHREC'17 Track dataset [61] is a public dynamic hand gesture dataset that provides depth image and skeleton data. Depth images of various lengths are acquired by an Intel RealSense camera SR300 [37], and hand skeletons are captured using the supplied software development kit (SDK). The dataset contains the following 14 gestures obtained from 28 right-handed participants: Grab, Tap, Expand, Pinch, Rotation clockwise (RC), Rotation counterclockwise (RCC), Swipe right (SR), Swipe left (SL), Swipe up (SU), Swipe down (SD), Swipe x (SX), Swipe + (S+), Swipe v (SV), and Shake. Differences in both hand movement and hand shape change must be considered to satisfy the requirement of an application available in a human-computer interface. Therefore, this dataset divides 14 gestures into two categories: fine and coarse. Fine gestures are motions that represent changes in the shape of the hand, and coarse gestures are motions related to the movement of the entire hand.

As shown in Fig. 2(b), each gesture is performed with two hand shapes of one finger and the whole hand—gestures are classified with either 14 or 28 labels, depending on the number of fingers used. The hand sequence is captured at 30 frames per second, and the maximum length of frames is 180. Each frame contains a depth image with 640×480 resolution and coordinates of 22 hand joint positions in the 3D world space and 2D image space. The structure of the skeleton returned by the camera is shown in Fig. 2(a). The dataset consists of 2,800 sequences, with 1,960 sequences in the training dataset and 840 in the testing dataset.

2) *DHG-14/28*: The DHG-14/28 dataset [17] uses the same data collection method as the SHREC'17 Track dataset and consists of 2800 sequences of 22 hand joints. Skeleton data are collected from 20 participants, and each movement is performed 5 times. The length of the sequence ranges from 20 to 150 frames. In contrast to the SHREC'17 Track dataset, the DHG-14/28 dataset uses a leave-one-subject-out experimental protocol. This is a cross-validation method in which

the data of 19 subjects are used as the training dataset and the remaining one subject is used as the testing dataset. This method is conducted 20 times, with the average used as the final recognition accuracy.

B. Training Details

We performed all our experiments with the PyTorch framework using two NVIDIA TITAN Xp GPUs. We applied stochastic gradient descent (SGD) for training with a Nesterov momentum of 0.9 as an optimization strategy. Furthermore, the weight decay was set to 0.0001. We adopted the cross-entropy loss as the loss function for the final prediction and minimized it. The batch size was 32 and the training was ended in 150 epochs. The initial learning rate was 0.1 and divided by 10 at 90 epochs and 130 epochs.

For stable initial learning, we applied a warm-up strategy [62] that linearly increases the learning rate from 0 to an initial value of 0.1 during the first 20 epochs. Moreover, a dropout layer with a drop rate of 0.4 was added between the GAP and FC layers. We disturbed the hand joint coordinates by applying random translation for data augmentation. Because each gesture has a different frame sequence length, we set the length of all sequences as the maximum number of frames to fully exploit the provided coordinate data. We set sequences with less than the maximum number of frames to zero until they reach the maximum. Therefore, the SHREC'17 Track dataset was set to 180, and the DHG-14/28 dataset was set to 150.

C. Ablation Study

The proposed MS-ISTGCN consists of various novel components. This section discusses the validity of the components in the proposed MS-ISTGCN on the SHREC'17 Track and action recognition datasets with only skeleton joint data. We found the optimal condition with the highest performance through an ablation study and applied it to the remaining experiments on both datasets.

1) *Comparisons of Block Architectures*: The dynamic hand gesture datasets have fewer frames and nodes and are smaller than the human action datasets. Therefore, we experimented with various architectures by changing the number of blocks to find the optimal structure for the dynamic hand gesture dataset. As presented in Table I, the architecture of the eight blocks had the highest accuracy. This architecture not only improved performance by reducing the overfitting of the network but also had the effect of reducing the size of the model.

2) *Comparisons of Adaptive Spatial GC Layers*: The proposed adaptive graph convolution layer can learn graph topology better adapted to the GCN-based dynamic hand gesture recognition. This adaptive GC layer consists of the sum of three adjacency matrices. We tested the performance of using each of the matrices and several combinations of matrices to find the optimal graph adjacency matrix, with the results presented in Table II. The experiment results reveal that the highest accuracy is achieved when applying the A+B+C adjacency matrix, which adds the original adjacency matrix A,

TABLE I

PERFORMANCE COMPARISONS OF DIFFERENT NUMBER OF ISTGCN BLOCKS ON SHREC'17 TRACK DATASET WITH 14 GESTURES IN ACCURACY AND PARAMETER NUMBER. 64B DENOTES THE "64-CHANNEL BLOCK"

Blocks	Accuracy (%)	Params (M)
6 (64B $\times$ 2, 128B $\times$ 2, 256B $\times$ 2)	95.00	1.68
7 (64B $\times$ 3, 128B $\times$ 2, 256B $\times$ 2)	95.71	1.72
8 (64B $\times$ 4, 128B $\times$ 2, 256B $\times$ 2)	95.83	1.77
9 (64B $\times$ 4, 128B $\times$ 3, 256B $\times$ 2)	94.88	1.93
10 (64B $\times$ 4, 128B $\times$ 3, 256B $\times$ 3)	94.88	2.58

TABLE II

PERFORMANCE COMPARISONS OF DIFFERENT SPATIAL ADJACENCY MATRIX ON SHREC'17 TRACK DATASET WITH 14 GESTURES IN ACCURACY AND PARAMETER NUMBER

Methods	Accuracy (%)	Params (M)
A	94.76	1.56
B	94.76	1.56
C	94.88	1.76
A+B (w/o C)	94.88	1.56
A+C (w/o B)	95.00	1.76
B+C (w/o A)	94.76	1.77
A+B+C	95.83	1.77

TABLE III

PERFORMANCE COMPARISONS OF DIFFERENT TEMPORAL GRAPH CONVOLUTION LAYER TYPES ON SHREC'17 TRACK DATASET WITH 14 GESTURES IN ACCURACY AND PARAMETER NUMBER. E DENOTES THE "EXTENDED" AND Γ DENOTES THE KERNEL SIZE OF $\Gamma \times 1$ CONVOLUTIONS DILATIONS OF THE E-TGC METHODS ARE 1, 2 AND 3

Methods	Accuracy (%)	Params (M)
Basic TGC, $\Gamma=9, d=1$	95.12	2.89
E-TGC, $\Gamma=3$	94.64	1.63
E-TGC, $\Gamma=5$	95.36	1.70
E-TGC, $\Gamma=7$	95.83	1.77
E-TGC, $\Gamma=9$	95.36	1.83

TABLE IV

PERFORMANCE COMPARISONS OF DIFFERENT ATTENTION LAYERS ON SHREC'17 TRACK DATASET WITH 14 GESTURES IN ACCURACY AND PARAMETER NUMBER

Methods	Accuracy (%)	Params (M)
w/o Att.	94.29	1.22
STC Att. [47]	94.76	1.43
ST-Joint Att. [45]	95.12	1.50
Eff-ST Att.	94.88	1.59
Eff-ST + Ch Att.	95.83	1.77

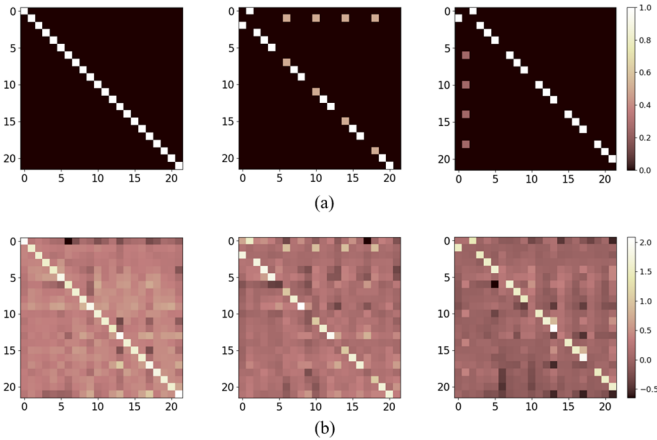


Fig. 8. Example of the visualized adjacency matrices (a) the original adjacency matrices and (b) the learned adjacency matrices of the first block.

parameterized matrix B, and data-dependent matrix C multiplied by α . This method improved approximately 1.1% compared with the one applying only the original adjacency matrix A. Fig. 8 illustrates the original adjacency matrices and the learned adjacency matrices. The three matrices of Fig. 8(a) were the original adjacency matrices connected as shown in Fig. 2(a), and the three matrices of Fig. 8(b) were the learned adjacency matrices in the first block. As shown in Fig. 8, the adaptively learned adjacency matrices could acquire latent relationships between joints without constraints on physical connections. Moreover, considering the sample-dependent adjacency matrix, the performance was further improved.

3) *Comparisons of Extended Temporal GC Layers:* We proposed an extended temporal graph convolution layer using multiple scale convolutions to obtain discriminative and informative temporal features. We selected the optimal extended temporal graph convolution (E-TGC) layer for skeleton-based dynamic hand gesture recognition by testing four different kernel-sized TGC layers ($d = 1, 2, 3$), including basic TGC ($\Gamma = 9, d = 1$). The results are presented in Table III. As presented in Table III, the E-TGC with kernel size $\Gamma = 7$ had the best performance. Furthermore, the proposed E-TGC layer was a low computational architecture with significantly fewer parameters than the basic TGC method because it used a bottleneck structure.

4) *Comparison of Attention Layers:* We evaluated the proposed attention layer consisting of two sub-layers in comparison with previous methods. We compared the proposed combination of effective spatio-temporal attention and channel attention with other attention layers, (spatial-temporal-channel (STC) attention [45] and ST-Joint attention [46]), with the results presented in Table IV. The model's accuracy improved significantly after using the attention layers compared with the method without the attention layer. Furthermore, our proposed method that applies effective-ST and channel attention together achieved much higher accuracy, although the model complexity (number of parameters) slightly increased compared with the other methods.

5) *Effect of Multi-Stream Network:* We verified the effectiveness of using three input modalities, presenting the results in Table V. We tested both 14 and 28 gestures for the SHREC'17 Track dataset, with multi-stream methods outperforming single-stream methods. The three single-stream

TABLE V
ACCURACY COMPARISONS OF THE DIFFERENT DATA STREAMS
AND COMBINATIONS ON SHREC'17 TRACK DATASET

Methods	Accuracy (%)	
	14 Gestures	28 Gestures
Joint	95.83	94.29
Bone	95.36	93.81
Motion	94.17	90.48
J + B	96.07	94.40
J + B + JM	96.67	94.88

TABLE VI
COMPARISONS OF THE ACTION RECOGNITION PERFORMANCE WITH
THE STATE-OF-THE-ART METHODS ON THE NTU
RGB+D SKELETON DATASET

Methods	Accuracy (%)		Params (M)
	X-Sub	X-View	
ST-GCN [43]	81.5	88.3	3.1
2s AGCN [44]	88.5	95.1	7.0
SGN [50]	89.0	94.5	0.7
4s DGNN [51]	89.9	96.1	26.2
4s MS-AAGCN [45]	90.0	96.2	15.1
2s MS-G3D [49]	91.5	96.2	6.4
4s Dynamic-GCN [52]	91.5	96.0	14.4
Ours (JB-ISTGCN)	90.6	96	3.6
Ours (3s ISTGCN)	91.3	96.3	5.3
Ours (4s ISTGCN)	91.4	96.4	7.1

methods exhibit high performance in the order of joint modality, bone modality, and motion modality. The combination of joint and bone modality achieved a slight improvement on the 14- and 28-gesture categories compared with the joint modality. The highest performance improvement occurred when fusing the three modalities. Motion modality was essential for improving performance in multi-stream, although single-stream performance was relatively low.

6) *Generalization Performance on Action Recognition:* Finally, we verified the generalization performance on action recognition using the NTU RGB+D dataset [63]. It is the most widely used action recognition dataset, which contains 56,578 skeleton sequences over 60 action classes captured from 40 volunteers and 3 cameras with different views. This dataset provides 25 body joints as nodes. It recommends two settings: cross-subject (X-Sub) and cross-view (X-View), where the subjects and cameras used in the training and test splits are different, respectively.

Table VI presents the performance comparison for the proposed and state-of-the-art methods on the action recognition dataset. All methods compared are based on GCN and are almost multi-stream structures. We further evaluated the four streams method with the addition of bone motion stream to compare with equally other methods on a large dataset. The proposed method shows comparable accuracy for X-Sub and the best accuracy for X-View, respectively. Therefore, the proposed method has good generalization performance on

Ground Truth	Grab	75.2	1.9	1.4	20.0	0.0	0.0	0.0	0.0	0.0	1.4	0.0	0.0	0.0	0.0
	Tap	4.3	93.3	0.0	1.0	0.5	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0
	Expand	0.0	0.0	97.1	1.0	0.0	0.0	0.5	0.0	1.0	0.5	0.0	0.0	0.0	0.0
	Pinch	20.5	2.9	0.5	74.3	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
	RC	0.5	1.4	0.0	1.4	92.9	0.0	0.5	2.9	0.0	0.0	0.0	0.0	0.0	0.5
	RCC	1.0	0.0	1.4	0.0	0.0	96.7	0.0	0.0	0.5	0.5	0.0	0.0	0.0	0.0
	SR	0.0	0.0	0.0	0.0	0.0	0.0	99.5	0.0	0.5	0.0	0.0	0.0	0.0	0.0
	SL	0.0	0.0	0.0	1.0	2.4	1.0	0.0	95.7	0.0	0.0	0.0	0.0	0.0	0.0
	SU	0.0	0.0	4.3	0.0	0.0	0.0	0.0	95.7	0.0	0.0	0.0	0.0	0.0	0.0
	SD	0.0	1.0	0.5	1.0	0.5	0.0	0.0	0.5	96.7	0.0	0.0	0.0	0.0	0.0
	SX	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.0	96.7	1.0	1.9	0.0	0.0
	S+	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	99.0	0.5	0.0	0.0
	SV	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.5	0.0	0.0	98.6	0.0	0.0
	Shake	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	99.0	0.0
		Grab	Tap	Expand	Pinch	RC	RCC	SR	SL	SU	SD	SX	S+	SV	Shake
		Predicted													

Fig. 9. Confusion matrix of the proposed method on SHREC'17 Track dataset with 14 gestures.

TABLE VII
COMPARISONS OF DYNAMIC HAND GESTURE RECOGNITION
PERFORMANCE WITH THE STATE-OF-THE-ART METHODS
ON SHREC'17 TRACK DATASET. MS DENOTES
THE "MULTI-STREAM"

Methods	Accuracy (%)	
	14 Gestures	28 Gestures
HON4D [21]	78.5	74.0
SoCJ + HoHD + HoWR [17]	88.2	81.9
MFA-Net [24]	91.3	86.6
ST-GCN [43]	92.7	87.7
HG-GCN [54]	92.8	88.3
STA-Res-TCN [35]	93.6	90.7
ST-TS-HGR-NET [32]	94.3	89.4
DG-STA [53]	94.4	90.7
DD-Net [30]	94.6	91.9
HPEV [33]	94.9	92.3
SPTP [34]	96.3	93.3
DLSTM [31]	97.6	91.4
SAGCN + RBi-IndRNN [25]	96.3	94.1
Ours (MS-ISTGCN)	96.7	94.9

action recognition as well as hand gesture recognition task. In addition, the proposed method has the advantage of relatively small computational cost compared to other methods.

D. Comparison With State-of-the-Art Methods

We compared the performance of the proposed method and several state-of-the-art methods on the SHREC'17 Track dataset and DHG-14/28 dataset, respectively. The results are as follows.

1) *Performance Comparison on SHREC'17 Track Dataset:* Table VII presents the results of comparing our method, hand-crafted feature-based methods [17, 21], and deep learning-based methods [24], [25], [30]–[35], [43], [53], [54] on the

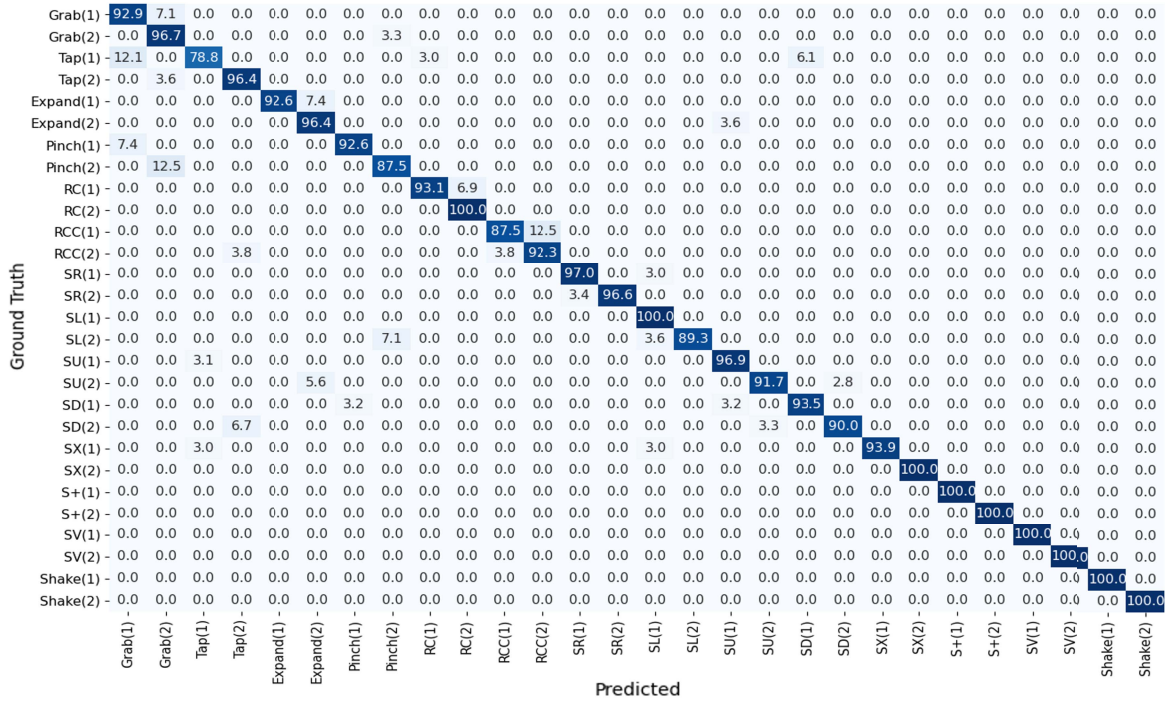


Fig. 10. Confusion matrix of the proposed method on SHREC'17 Track dataset with 28 gestures. The gestures annotated (1) and (2) are respectively performed using one finger and with the whole hand.

SHREC'17 Track dataset. The accuracy of the DLSTM [31] method was 97.6% for the 14-gesture category. This method used handcrafted angular features from hand joints as the input to the LSTM-RNN network. The accuracies of the SAGCN + RBi-IndRNN [25] method were 96.3% and 94.1% for the 14- and 28-gesture categories, respectively. This method was a two-stream network fusing a self-attention-based GCN that is advantageous for capturing spatial information and a bi-directional RNN, which is advantageous for capturing temporal information. However, these RNN-based networks still had a gradient decay problem during training, resulting in performance degradation.

Our method achieved the second-highest recognition accuracy of 96.7% on 14 gestures, which is 0.9% lower than the DLSTM [31] method. The accuracy of the proposed network on 28 gestures is 94.9%, which is 0.8% higher than the state-of-the-art method because our method used enhanced attention features that incorporate both learned spatial information and temporal information captured from a wide receptive field, fusing scores from the three modalities to predict gestures. Therefore, the proposed method outperformed these conventional approaches and achieved the competitive and best performance.

Figs. 9 and 10 show the confusion matrices on 14 and 28 gestures for the SHREC'17 Track dataset. The color of the confusion matrix indicates prediction accuracy while distinguishing each label. Our method achieved 100% accuracy in specific categories. Furthermore, the recognition rate on 14 gestures was over 88%, exhibiting high performance. As shown in Fig. 10, the relatively low accuracy of the "Tab" gesture was due to the motion similarity between "Tab" and "Grab" in the one finger category.

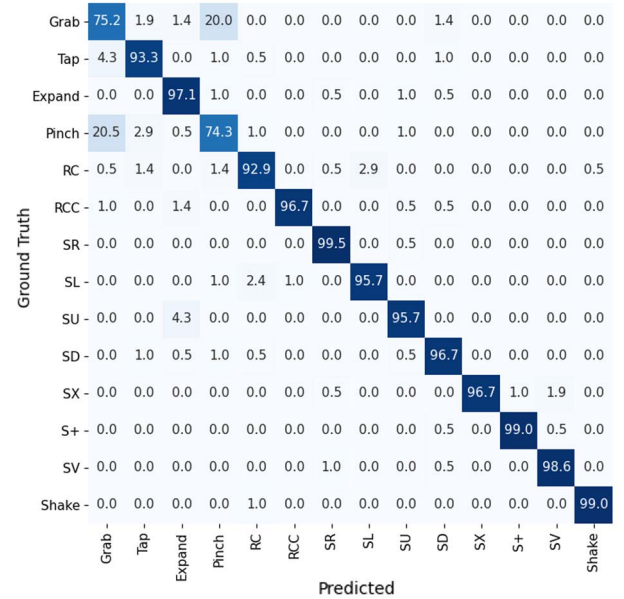
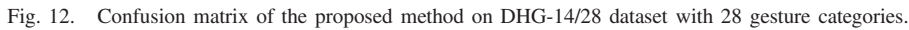


Fig. 11. Confusion matrix of the proposed method on DHG-14/28 dataset with 14 gesture categories.

2) *Performance Comparison on DHG-14/28 Dataset:* Table VIII presents the performance comparison for the proposed and state-of-the-art methods on the DHG-14/28 dataset. Our method outperformed the state-of-the-art methods on both the 14- and 28-gesture categories of the DHG-14/28 dataset and achieved accuracies of 93.7% and 91.2%, respectively. The spatial perception and temporal perception streams (SPTP) [34] method maximized the advantages of 2D-CNN



V. CONCLUSION AND FUTURE RESEARCH

This paper proposed an MS-ISTGCN with a spatial GC layer, enhanced attention layer, and improved temporal GC layer for skeleton-based dynamic hand gesture recognition. Moreover, an adaptive graph convolution layer that can learn the relationship between distant hand joints and a multiple scale temporal convolution layer that can cover a wide receptive field was proposed to model spatial pose features and temporal motion features of dynamic hand gestures. An attention layer was also proposed that uses a combination of effective spatio-temporal co-attention and channel attention to find the most informative joints, frames, and features from skeleton sequences. Finally, performance was improved by applying a multi-stream structure with joints, bones, and joint motions as inputs to take full advantage of the low-level features. Experimental results demonstrated that our network outperforms the state-of-the-art methods on two public datasets of dynamic hand gesture recognition.

Future research will focus on how to combine a 3D hand pose estimation algorithm and a skeleton-based dynamic hand gesture recognition algorithm in one framework.

REFERENCES

- [1] S. S. Rautaray and A. Agrawal, "Vision based hand gesture recognition for human computer interaction: A survey," *Artif. Intell. Rev.*, vol. 43, no. 1, pp. 1–54, Jan. 2015.
- [2] H. Cheng, L. Yang, and Z. Liu, "Survey on 3D hand gesture recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 26, no. 9, pp. 1659–1673, Sep. 2016.
- [3] N. C. Camgoz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, "Neural sign language translation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 93–7784.
- [4] C. Linqin, C. Shuangjie, and X. Min, "Dynamic hand gesture recognition using RGB-D data for natural human-computer interaction," *J. Intell. Fuzzy Syst.*, vol. 323, no. 3, pp. 495–507, 2017.

- [5] T.-D. Tan and Z.-M. Guo, "Research of hand positioning and gesture recognition based on binocular vision," in *Proc. IEEE Int. Symp. VR Innov.*, Mar. 2011, pp. 311–315.
- [6] Y. Zhang, C. Cao, J. Cheng, and H. Lu, "Egogesture: A new dataset and benchmark for egocentric hand gesture recognition," *IEEE Trans. Multimedia*, vol. 20, no. 5, pp. 1038–1050, May 2018.
- [7] J. L. Raheja, G. A. Rajsekhar, and A. Chaudhary, "Controlling a remotely located robot using hand gestures in real time: A DSP implementation," in *Proc. 5th Int. Conf. Wireless Netw. Embedded Syst. (WECON)*, Oct. 2016, pp. 1–5.
- [8] R. Li, H. Wang, and Z. Liu, "Survey on mapping human hand motion to robotic hands for teleoperation," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Aug. 8, 2021, doi: [10.1109/TCSVT.2021.3057992](https://doi.org/10.1109/TCSVT.2021.3057992).
- [9] G.-F. He, J.-W. Park, S.-K. Kang, and S.-T. Jung, "Development of gesture recognition-based serious games," in *Proc. IEEE-EMBS Int. Conf. Biomed. Health Informat.*, Jan. 2012, pp. 922–925.
- [10] M. Oudah, A. Al-Naji, and J. Chahl, "Hand gesture recognition based on computer vision: A review of techniques," *J. Imag.*, vol. 6, no. 8, p. 73, Jul. 2020.
- [11] L. Dipietro, A. M. Sabatini, and P. Dario, "A survey of glove-based systems and their applications," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 38, no. 4, pp. 461–482, Jul. 2008.
- [12] W. T. Freeman and M. Roth, "Orientation histograms for hand gesture recognition," in *Proc. Int. Workshop Automatic Face Gesture Recognit.*, Jun. 1995, pp. 296–301.
- [13] P. Molchanov, S. Gupta, K. Kim, and J. Kautz, "Hand gesture recognition with 3D convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 1–7.
- [14] C. Wang, Z. Liu, and S.-C. Chan, "Superpixel-based hand gesture recognition with Kinect depth camera," *IEEE Trans. Multimedia*, vol. 17, no. 1, pp. 29–39, Jan. 2015.
- [15] P. Narayana, J. R. Beveridge, and B. A. Draper, "Gesture recognition: Focus on the hands," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5235–5244.
- [16] J. Sanchez-Riera, K. Srinivasan, K.-L. Hua, W.-H. Cheng, M. A. Hossain, and M. F. Alhamid, "Robust RGB-D hand tracking using deep learning priors," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 9, pp. 2289–2301, Sep. 2018.
- [17] Q. De Smedt, H. Wannous, and J.-P. Vandeborrel, "Skeleton-based dynamic hand gesture recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 1206–1214.
- [18] Q. De Smedt, H. Wannous, and J.-P. Vandeborrel, "Heterogeneous hand gesture recognition using 3D dynamic skeletal data," *Comput. Vis. Image Understand.*, vol. 181, pp. 60–72, Apr. 2019.
- [19] S. Y. Boulahia, E. Anquetil, F. Multon, and R. Kulpa, "Dynamic hand gesture recognition based on 3D pattern assembled trajectories," in *Proc. 7th Int. Conf. Image Process. Theory, Tools Appl. (IPTA)*, Nov. 2017, pp. 1–6.
- [20] F. M. Caputo, P. Prebianca, A. Carcangiu, L. D. Spano, and A. Giachetti, "Comparing 3D trajectories for simple mid-air gesture recognition," *Comput. Graph.*, vol. 73, pp. 17–25, Jun. 2018.
- [21] O. Oreifej and Z. Liu, "HON4D: Histogram of oriented 4D normals for activity recognition from depth sequences," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 716–723.
- [22] X. Chen, H. Guo, G. Wang, and L. Zhang, "Motion feature augmented recurrent neural network for skeleton-based dynamic hand gesture recognition," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 2881–2885.
- [23] L. Zhang, G. Zhu, P. Shen, J. Song, S. A. Shah, and M. Bennamoun, "Learning spatiotemporal features using 3DCNN and convolutional LSTM for gesture recognition," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 3120–3128.
- [24] X. Chen, G. Wang, H. Guo, C. Zhang, H. Wang, and L. Zhang, "MFA-net: Motion feature augmented network for dynamic hand gesture recognition from skeletal data," *Sensors*, vol. 19, no. 2, p. 239, Jan. 2019.
- [25] C. Li, S. Li, Y. Gao, X. Zhang, and W. Li, "A two-stream neural network for pose-based hand gesture recognition," 2021, *arXiv:2101.08926*.
- [26] J. Weng, M. Liu, X. Jiang, and J. Yuan, "Deformable pose traversal convolution for 3D action and gesture recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 142–157.
- [27] J. C. Núñez, R. Cabido, J. J. Pantrigo, A. S. Montemayor, and J. F. Vélez, "Convolutional neural networks and long short-term memory for skeleton-based human activity and hand gesture recognition," *Pattern Recognit.*, vol. 76, pp. 80–94, Apr. 2018.
- [28] S. Shin and W.-Y. Kim, "Skeleton-based dynamic hand gesture recognition using a part-based GRU-RNN for gesture-based interface," *IEEE Access*, vol. 8, pp. 50236–50243, 2020.
- [29] G. Devineau, F. Moutarde, W. Xi, and J. Yang, "Deep learning for hand gesture recognition on skeletal data," in *Proc. 13th IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, May 2018, pp. 106–113.
- [30] F. Yang, Y. Wu, S. Sakti, and S. Nakamura, "Make skeleton-based action recognition model smaller, faster and better," in *Proc. ACM Multimedia Asia*, Dec. 2019, pp. 1–6.
- [31] D. Avola, M. Bernardi, L. Cinque, G. L. Foresti, and C. Massaroni, "Exploiting recurrent neural networks and leap motion controller for the recognition of sign language and semaphoric hand gestures," *IEEE Trans. Multimedia*, vol. 21, no. 1, pp. 234–245, Jan. 2019.
- [32] X. S. Nguyen, L. Brun, O. Lezorey, and S. Bougleux, "A neural network based on SPD manifold learning for skeleton-based hand gesture recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, p. 12036.
- [33] J. Liu, Y. Liu, Y. Wang, V. Prinet, S. Xiang, and C. Pan, "Decoupled representation learning for skeleton-based gesture recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5751–5760.
- [34] Y. Li, D. Ma, Y. Yu, G. Wei, and Y. Zhou, "Compact joints encoding for skeleton-based dynamic hand gesture recognition," *Comput. Graph.*, vol. 97, pp. 191–199, Jun. 2021.
- [35] J. Hou, G. Wang, X. Chen, J.-H. Xue, R. Zhu, and H. Yang, "Spatial-temporal attention res-TCN for skeleton-based dynamic hand gesture recognition," in *Proc. Eur. Conf. Comput. Vis. Workshops*, Sep. 2018, pp. 273–286.
- [36] Z. Zhang, "Microsoft Kinect sensor and its effect," *IEEE MultiMedia*, vol. 19, no. 2, pp. 4–10, Feb. 2012.
- [37] L. Keselman, J. I. Woodfill, A. Grunnet-Jepsen, and A. Bhowmik, "Intel(R) RealSense(TM) stereoscopic depth cameras," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1–10.
- [38] J.-H. Song and S.-J. Kang, "3D hand pose estimation via graph-based reasoning," *IEEE Access*, vol. 9, pp. 35824–35833, 2021.
- [39] P. Ren et al., "Spatial-aware stacked regression network for real-time 3D hand pose estimation," *Neurocomputing*, vol. 437, pp. 42–57, May 2021.
- [40] M. Oberweger and V. Lepetit, "DeepPrior++: Improving fast and accurate 3D hand pose estimation," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 585–594.
- [41] H. Liang, J. Yuan, and D. Thalmann, "Resolving ambiguous hand pose predictions by exploiting part correlations," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 7, pp. 1125–1139, Jul. 2015.
- [42] L. Xu, C. Hu, J. Tao, J. Xue, and K. Mei, "Improve regression network on depth hand pose estimation with auxiliary variable," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 890–904, Mar. 2021.
- [43] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 7444–7452.
- [44] L. Shi, Y. Zhang, J. Cheng, and H. Lu, "Two-stream adaptive graph convolutional networks for skeleton-based action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12026–12035.
- [45] L. Shi, Y. Zhang, J. Cheng, and H. Lu, "Skeleton-based action recognition with multi-stream adaptive graph convolutional networks," *IEEE Trans. Image Process.*, vol. 29, pp. 9532–9545, 2020.
- [46] Y.-F. Song, Z. Zhang, C. Shan, and L. Wang, "Constructing stronger and faster baselines for skeleton-based action recognition," 2021, *arXiv:2106.15125*.
- [47] C. Wu, X.-J. Wu, and J. Kittler, "Graph2Net: Perceptually-enriched graph learning for skeleton-based action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Jun. 3, 2021, doi: [10.1109/TCSVT.2021.3085959](https://doi.org/10.1109/TCSVT.2021.3085959).
- [48] Y.-F. Song, Z. Zhang, C. Shan, and L. Wang, "Richly activated graph convolutional network for robust skeleton-based action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 5, pp. 1915–1925, May 2021.
- [49] Z. Liu, H. Zhang, Z. Chen, Z. Wang, and W. Ouyang, "Disentangling and unifying graph convolutions for skeleton-based action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 143–152.
- [50] P. Zhang, C. Lan, W. Zeng, J. Xing, J. Xue, and N. Zheng, "Semantics-guided neural networks for efficient skeleton-based human action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1112–1121.

- [51] L. Shi, Y. Zhang, J. Cheng, and H. Lu, "Skeleton-based action recognition with directed graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7912–7921.
- [52] F. Ye, S. Pu, Q. Zhong, C. Li, D. Xie, and H. Tang, "Dynamic GCN: Context-enriched topology learning for skeleton-based action recognition," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 55–63.
- [53] Y. Chen, L. Zhao, X. Peng, J. Yuan, and D. N. Metaxas, "Construct dynamic graphs for hand gesture recognition via spatial-temporal attention," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, 2019, pp. 1–13.
- [54] Y. Li, Z. He, X. Ye, Z. He, and K. Han, "Spatial temporal graph convolutional networks for skeleton-based dynamic hand gesture recognition," *Eurasip J. Image Video Process.*, vol. 2019, no. 1, pp. 1–7, 2019.
- [55] M. Tan *et al.*, "Bidirectional posture-appearance interaction network for driver behavior recognition," *IEEE Trans. Intell. Transp. Syst.*, early access, Nov. 13, 2021, doi: [10.1109/TITS.2021.3123127](https://doi.org/10.1109/TITS.2021.3123127).
- [56] C. Si, W. Chen, W. Wang, L. Wang, and T. Tan, "An attention enhanced graph convolutional LSTM network for skeleton-based action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1227–1236.
- [57] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017, pp. 1–14.
- [58] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4 inception-ResNet and the impact of residual connections on learning," in *Proc. AAAI*, 2016, pp. 4278–4284.
- [59] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
- [60] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13713–13722.
- [61] Q. D. Smedt, H. Wannous, J.-P. Vandeboer, J. Guerry, B. L. Saux, and D. Filliat, "SHREC'17 track: 3D hand gesture recognition using a depth and skeletal dataset," in *Proc. Eurographics Workshop 3D Object Retr.*, 2017, pp. 33–38.
- [62] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [63] A. Shahroudy, J. Liu, T.-T. Ng, and G. Wang, "NTU RGB+D: A large scale dataset for 3D human activity analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1010–1019.



Jae-Hun Song received the B.S. and M.S. degrees in electronic engineering from Hanyang University, South Korea, in 2001 and 2003, respectively. He is currently pursuing the Ph.D. degree in electronic engineering with Sogang University. He is working in circuit design and medium display product development with LG Display Company, Ltd., South Korea. His current research interests include image processing, computer vision, and deep learning.



Kyeongbo Kong received the B.S. degree in electronics engineering from Sogang University, Seoul, South Korea, in 2015, and the M.S. and Ph.D. degrees in electrical engineering from the Pohang University of Science and Technology (POSTECH), Pohang, South Korea, in 2017 and 2020, respectively. From 2020 to 2021, he was worked as a Post-Doctoral Fellow at the Department of Electrical Engineering, POSTECH. He is currently an Assistant Professor with the Department of Media School, Pukyong National University, Busan, South Korea.

His current research interests include image processing, computer vision, machine learning, and deep learning.



Suk-Ju Kang (Member, IEEE) received the B.S. degree in electronic engineering from Sogang University, South Korea, in 2006, and the Ph.D. degree in electrical and computer engineering from the Pohang University of Science and Technology, in 2011. From 2011 to 2012, he was a Senior Researcher at LG Display, where he was a Project Leader for resolution enhancement and multi-view 3D system projects. From 2012 to 2015, he was an Assistant Professor of electrical engineering at Dong-A University, Busan. He is currently a Professor of electronic engineering with Sogang University. His current research interests include image analysis and enhancement, video processing, multimedia signal processing, circuit design for display systems, and deep learning systems. He was a recipient of the IEIE/IEEE Joint Award for Young IT Engineer of the Year, in 2019.